13 Three solid shapes **A**, **B** and **C** are similar.

The volume of shape **A** is $27\,cm^3$
The volume of shape **B** is $64\,cm^3$

The ratio of the surface area of shape **B** to the surface area of shape **C** is 8 : 15

Work out the ratio of the height of shape **A** to the height of shape **C**.

Volumes A : B = 27 : 64

Lengths A : B = 3 : 4

1 mark
Cube-root volume ratio.

Areas A : B = 9 : 16

1 mark
Square length ratio.

Areas B : C = 8 : 15 = 16 : 30

1 mark
Match the B areas.

So areas A : C = 9 : 30

Heights A : C = √9 : √30 = 3 : √30

1 mark
Square-root the area ratio to return to lengths.

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(Total for Question 13 is 4 marks)

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